

Midterm 1

ORF 309

Name: _____

Instructions: This midterm has 5 questions for a total of 100 points. Partial credit will be awarded for partial proofs or arguments. You may freely use basic facts from class, such as laws of probability, properties of conditional expectation, or manipulations with pmfs/pdfs. However, students must provide adequate justifications for especially important steps. Put another way, try your best to demonstrate awareness that you are applying the correct facts in the correct way by explicitly mentioning the results you are applying, indicating the objects they are applied to, and checking the required assumptions.

Good Luck!

Question	Points	Score
1	25	
2	20	
3	5	
4	30	
5	20	
Total:	100	

1. (Drawing Cards) Suppose you successfully draw cards independently and uniformly at random from a standard deck of 52 cards, i.e., 4 suits (hearts, diamonds, spades, and clubs) each with 13 denominations (labelled $A, 2, \dots, 10, J, Q, K$).
 - (a) (5 points) How many draws do you expect to perform before arriving at a heart if successive draws are done *with replacement*?
 - (b) (10 points) How many draws do you expect to perform before arriving at a heart if successive draws are done *without replacement*?
 - (c) (10 points) What is the probability of getting an ace (A) before a face (J, Q, K) if successive draws are done *with replacement*? How about *without replacement*? (Hint: Intuitive reasoning for the answers here is sufficient.)

2. (Trick Coins) Determine a blank coin by painting each side heads or tails with probability $1/2$ independently for each side.

- (a) (10 points) Given you flip heads with the painted coin, what is the probability your next flip is also heads?

Suppose the painting procedure above produced a double-headed coin, which gets comingled with 999 other fair coins. Suppose you pick a coin uniformly at random from the 1000 total and flip 10 heads in a row with the chosen coin.

- (a) (10 points) What is the probability you chose the painted double-headed coin?

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3. (Lottery) You enter a game that requires you to draw two ping-pong balls from a bag containing four that are labelled 1, 2, 3, 4. If at least one of the two you draw is strictly greater than 2, you win \$1; otherwise, you lose \$1. Let X be your net profit and let Y be the number on the first ball drawn.
- (a) (5 points) Compute $\mathbb{E}[X|Y]$ (intuitive reasoning for the expression is sufficient).

4. (Poisson processes)

Let $(N_t)_{t \geq 0}$ be a Poisson process of rate $\lambda > 0$, and let $\tau \sim \text{Exp}(\rho)$ be an independent exponential random variable of parameter ρ .

(a) (10 points) Suppose $X \sim \text{Exp}(\mu)$ where $\mu := \lambda + \rho$. Compute $\mathbb{E}[X^k]$ for $k \in \mathbb{N}$.

(b) (10 points) Compute $\mathbb{P}(N_\tau = k)$ for all $k \geq 0$.

Suppose the probability of observing at least one car during any 20 minutes is $\frac{609}{625}$. Assume arrivals occur with equal probability and independently “at every time.”

(a) (10 points) What is the probability of seeing at least one car during any 5 minute period?

5. (Rock, paper, scissors) This classic two-player game is played as follows. There are three different hand gestures representing rock, paper, or scissors. In every round of the game, each player simultaneously makes one of the gestures (so neither player knows what the other is going to choose).

If both players make the same gesture, the round is a tie. Otherwise, one of the players wins according to the following rules: **r** wins against **s**, **p** wins against **r**, and **s** wins against **p**. When the players tie, they keep playing additional rounds until one of them wins, at which point the game ends.

As each player cannot know what the other will play, the best they can do is the following strategy: in each round, each player independently chooses one of the gestures **r**, **p**, **s** with equal probability (that is, each player and each round is independent).

- (a) (5 points) What is the probability that the first round is a tie?
- (b) (5 points) What is the expected number N of rounds played until someone wins?
- (c) (5 points) What is the probability that the first player eventually wins the game?
- (d) (5 points) What is the expected number of times the first player plays **r** throughout the game?